EEET2601 Engineering Computing 1

Introduction

RMIT UNIVERSITY

What is a computer?





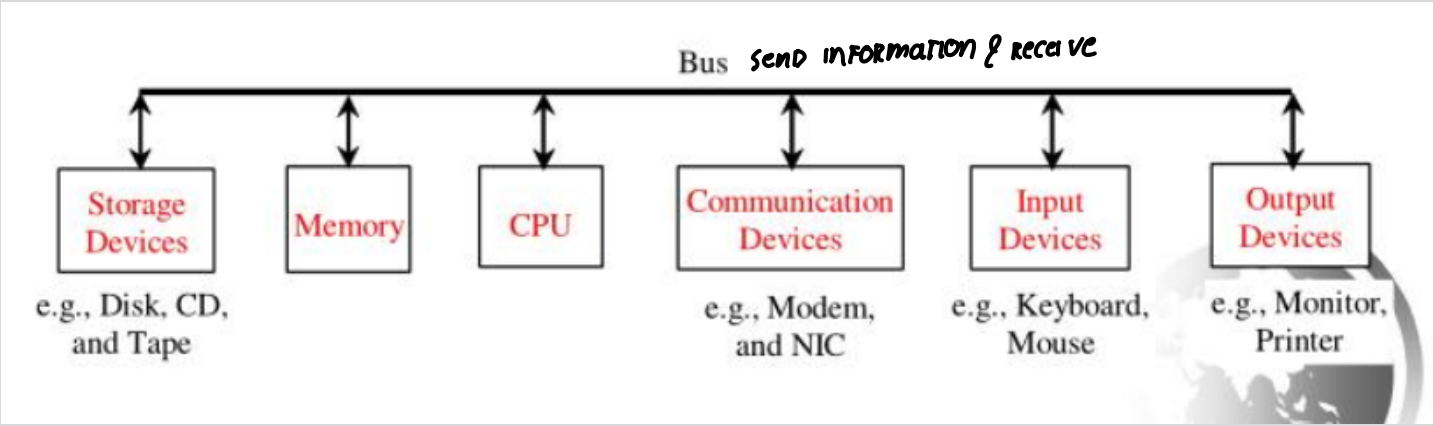






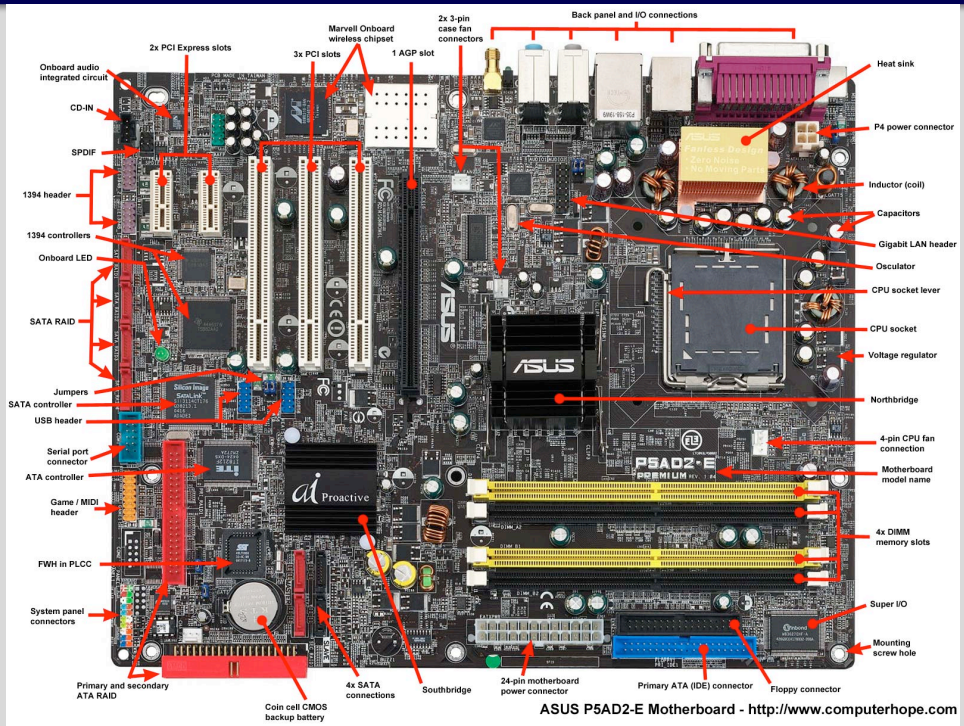
What is a computer?

A computer is a machine or device that can manipulate data based on the instructions provided by a program.



Motherboard

A motherboard holds and allows communication between the components of a computer.



Central Processing Unit (CPU)

- The CPU is the brain of a computer. It retrieves instructions from memory and execute them.
- Moore's Law: CPU speed will double every two years.
- A laptop today can have a CPU as an Intel Core i7 processor at 4.0 GHz (1 MHz = 1 million pulses per second).

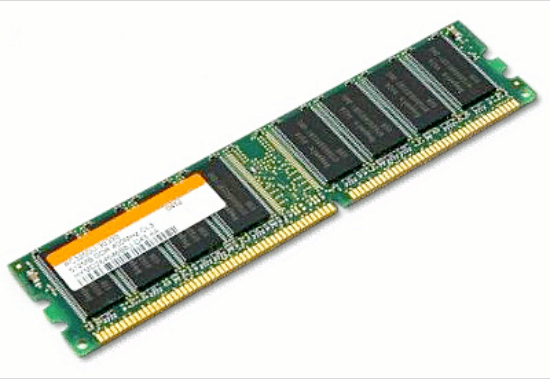
HOW IS IT AFECT THE PROCESSING TIME?



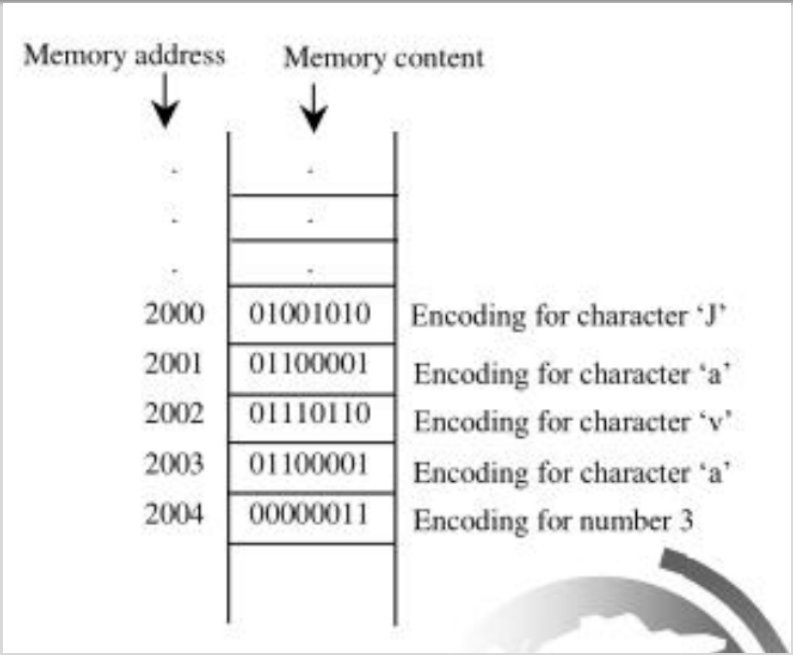
VS MHZ

Memory

- Often called RAM (Random Access Memory).
- RAM stores program instructions and data for the CPU to execute.
- RAM is an ordered sequence of bytes, each byte holds eight bits.
- A program and its data must be brought to RAM before they can be executed.

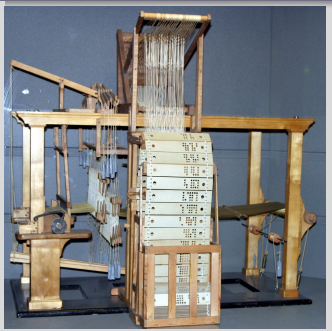


How data is stored?

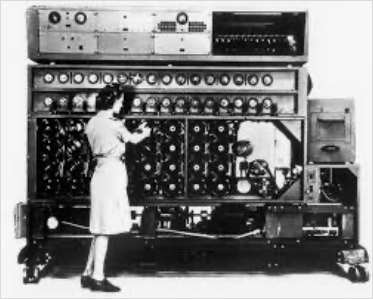


History of computers

- 1801: Joseph Jacquard invented a loom that uses punched wooden cards to automatically weave fabric design.
- 1822 – 1834: Charles Babbage invented the Difference Engine - the first mechanical computer - and then the Analytical Engine.
- 1890: Herman Hollerith designed a punch card system to calculate the 1880 census.
- 1939 – 1945: The bombe was an early electro-mechanical computer used by the British cryptologists to help decipher German Enigma-machine-encrypted secret messages during World War II.





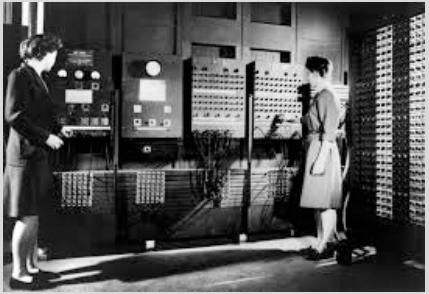




History of computers

- 1937: John Atanasoff and Clifford Berry built the first electronic digital computer. It was called ABC (Atanasoff Berry Computer).
- 1943 - 1944: John Mauchly and J. Presper Eckert built the ENIAC – the grandfather of digital computers. It filled a 20-foot by 40-foot room and has 18,000 vacuum tubes.
- 1945: Mauchly and Eckert went on to build UNIVAC – the first commercial computer for business and government applications.
- 1953: IBM made their mark in the computer world by introduced the 650 and 700 series computers.









History of computers

- 1981: IBM introduced Acorn - the first personal computer (PC) for home and office use.
- 1984: Apple invented Lisa - the Macintosh computer with a GUI.
- 1985: Toshiba launched Toshiba T1100 – the world's first mass-market laptop.
- 2007: Apple introduced iPhone which brings many computer functions to the smartphone.
- 2010: Apple unveiled iPad, which changes the way consumers view media and jumpstarting the dormant tablet computer segment.
- 2015: Apple released the Apple Watch.
- 2016: The first reprogrammable quantum computer was created.
- 2017: The Defense Advanced Research Project Agency (DARPA) is developing a new "Molecular Informatics" program that uses molecules as computers.













How's about these? COMPUTER INSIDE THE DEVICE

















Problem solving skill

- One of the most important skills for today's knowledge workers is **problem solving skill** - the ability to
  - **formulate** the problem clearly
  - **think creatively** to come up with a solution
  - **express** that solution precisely and concisely => algorithm

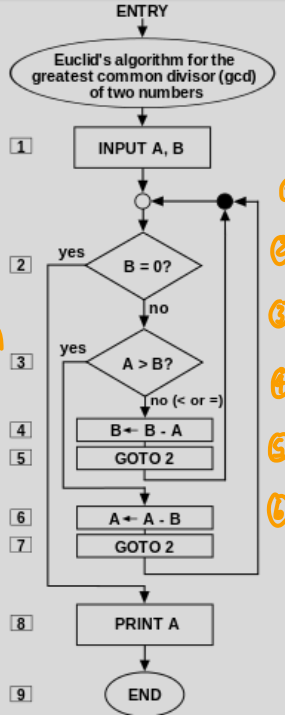
- An **algorithm** is a **sequence of instructions** that will solve the problem under consideration if followed exactly.

Question

What is the Greatest Common Divisor (GCD) of 12 and 15?

An example of an algorithm

- Euclid's algorithm for calculating the Greatest Common Divisor (GCD) of two numbers *a* and *b* in locations named A and B
- How well does this algorithm perform when A and B are about the same size or if they are very different sizes? BIG DIF → take longer
- Do you have any other ways you can determine the GCD of A and B? GCD IF 2# even (A/2 ; B/2) → X2 TO Have ans
- Do you know where the word algorithm comes from?



Challenge

You have a *3-liter* and a *5-liter* water container. Each container has no markings except for that which gives you its total volume. You also have a running tap. How to you use the containers and the tap in such a way to exactly measure out *4 liters* of water?

X, Y, Z

WHERE X = 3
Y = 5
Z = 4

IF (Z > X + Y) IMPOSSIBLE

IF [Z % GCD(X, Y) != 0] IMPOSSIBLE

else: POSSIBLE

1. FILL Y=5, X=0
2. POUR Y→X
   Y=2 X=3
3. EMTY X=0
4. POUR Y→X=2
   Y=0
5. FILL Y=5
   POUR TO X=2
   ↓
   Y=4 X=3

Why programming languages?

- An algorithm must be implemented as a program to run in a computer.
- A computer can only run programs written in its native low-level language ("machine language" or "assembly language").
- But writing programs in a low-level languages is very difficult.
- High-level languages were invented as they are
  - shorter
  - easier to read
  - taking less time to write
  - more *likely* to be correct
  - portable
  - closer to human languages

```
; Example of IBM PC assembly language
; Accepts a number in register AX;
; subtracts 32 if it is in the range 97-122;
; otherwise leaves it unchanged.

SUB32  PROC         ; procedure begins here
       CMP  AX,97   ; compare AX to 97
       JL   DONE    ; if less, jump to DONE
       CMP  AX,122  ; compare AX to 122
       JG   DONE    ; if greater, jump to DONE
       SUB  AX,32   ; subtract 32 from AX
DONE:  RET          ; return to main program
SUB32  ENDP         ; procedure ends here
```

FIGURE 17. Assembly language

Low-level vs high-level language

```
MONITOR FOR 6802 1.4          9-14-80  TSC ASSEMBLER  PAGE    2


C000                    ORG     ROM+$0000 BEGIN MONITOR
C000 8E 00 70  START    LDS     #STACK

               ****************************************
               * FUNCTION: INITA - Initialize ACIA
               * INPUT: none
               * OUTPUT: none
               * CALLS: none
               * DESTROYS: acc A

0013           RESETA  EQU     %00010011
0011           CTLREG  EQU     %00010001

C003 86 13     INITA   LDA A   #RESETA   RESET ACIA
C005 B7 80 04          STA A   ACIA
C008 86 11             LDA A   #CTLREG   SET 8 BITS AND 2 STOP
C00A B7 80 04          STA A   ACIA

C00D 7E C0 F1          JMP     SIGNON    GO TO START OF MONITOR

               ****************************************
               * FUNCTION: INCH - Input character
               * INPUT: none
               * OUTPUT: char in acc A
               * DESTROYS: acc A
               * CALLS: none
               * DESCRIPTION: Gets 1 character from terminal

C010 B6 80 04  INCH     LDA A   ACIA      GET STATUS
C013 47                 ASR A             SHIFT RDRF FLAG INTO CARRY
C014 24 FA              BCC     INCH      RECIEVE NOT READY
C016 B6 80 05           LDA A   ACIA+1    GET CHAR
C019 84 7F              AND A   #$7F      MASK PARITY
C01B 7E C0 79           JMP     OUTCH     ECHO & RTS

               ****************************************
               * FUNCTION: INHEX - INPUT HEX DIGIT
               * INPUT: none
               * OUTPUT: Digit in acc A
               * CALLS: INCH
               * DESTROYS: acc A
               * Returns to monitor if not HEX input

C01E 8D F0     INHEX   BSR     INCH      GET A CHAR
C020 81 30             CMP A   #'0       ZERO
C022 2B 11             BMI     HEXERR    NOT HEX
C024 81 39             CMP A   #'9       NINE
C026 2F 0A             BLE     HEXRTS    GOOD HEX
C028 81 41             CMP A   #'A
C02A 2B 09             BMI     HEXERR    NOT HEX
C02C 81 46             CMP A   #'F
C02E 2E 05             BGT     HEXERR
C030 80 07             SUB A   #7        FIX A-F
C032 84 0F     HEXRTS  AND A   #$0F      CONVERT ASCII TO DIGIT
C034 39                RTS

C035 7E C0 AF  HEXERR  JMP     CTRL      RETURN TO CONTROL LOOP
```

Motorola 6800

```
5 PRINT "Type two integers greater than 0"
10 INPUT A, B
20 IF B=0 THEN GOTO 80
30 IF A > B THEN GOTO 60
40 LET B=B-A
50 GOTO 20
60 LET A=A-B
70 GOTO 20
80 PRINT A
90 END
```

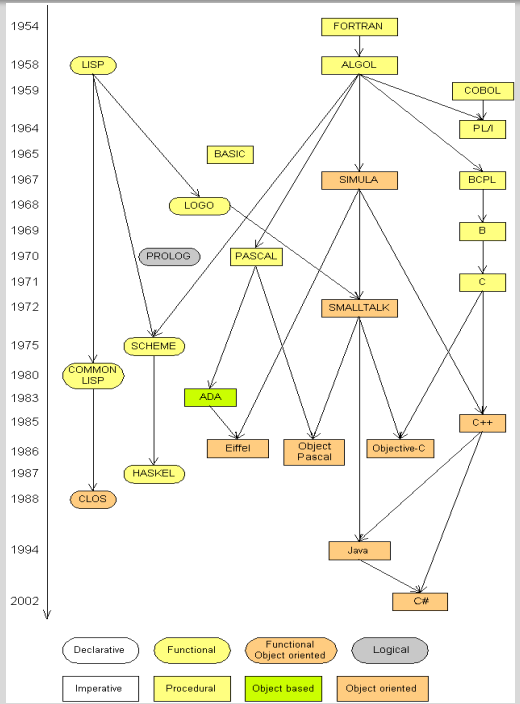
Fortran

```
int a, b;
printf("Type two integers greater than 0: ");
scanf("%i %i", &a, &b);
while (b != 0) {
    if (a > b) {
        a = a – b;
    } else {
        b = b – a;
    }
}
printf("%i", a);
```

C

Example of programming languages

- Some early influential ones:
  - FORTRAN - science / engineering
  - COBOL - business data
  - LISP - logic and AI
  - BASIC - a simple language
  - Pascal - education
- More modern ones include
  - C, C++, C#, Java, PHP, Perl, Ruby, Python, Swift, Go, etc.



What is a program?

- A program is a **sequence of instructions** written for the computer to follow in order to solve a particular problem
- A program is written in a programming language such as C, C++, Java, Python, etc.
- Main types of instructions:
  1. Input
  2. Output
  3. Compute
  4. Control

About C programming language

- Designed and developed by Dennis Ritchie at Bell Labs in 1973 to re-implement the UNIX operating system which was written early in assembly language.
- Fast, general-purpose, procedural, imperative.
- Many other programming languages inspired by the syntax and concepts from C. Can run on many devices | Fast small requirement for space
- According to the TIOBE Index, C is the most popular language as of Oct 2020.
- Most widely used programming language at all time.

Questions

Ada Lovelace

1. Who was the first programmer? And what was the first use of programming? Weaver in dustry lace machine
2. Why should engineers master programming?

easier to express idea to action

References

1. A. K. Zimmermann , History of Computers: A Brief Timeline, 2017.
2. D. Y. Liang, Intro to Java Programming, 10th edition, 2015.